



Analyzing and Forecasting Household Electricity Consumption Using Statistical, Machine Learning, and Deep Learning Approaches

Presentation by Fuel2Future group



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Research Rationale

- It is obvious to clarify that electricity plays an essential role in modern life and the socio-economic system.
- For a regulatory agency and policy planner, identifying the significant effects of weather and time on the level of electricity consumption, allowing for creating policy to coordinate the electricity supply by season or days.

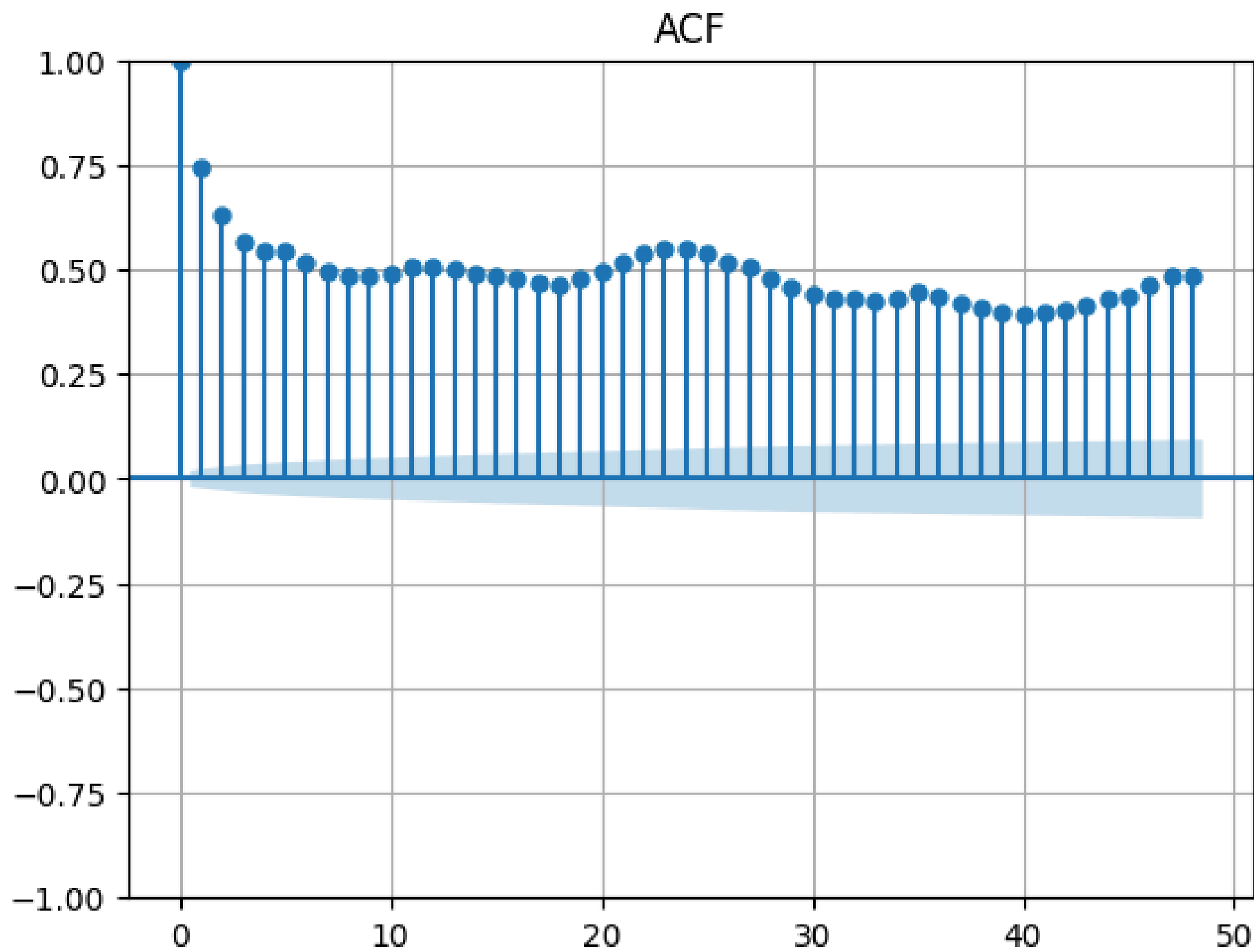


ARIMA

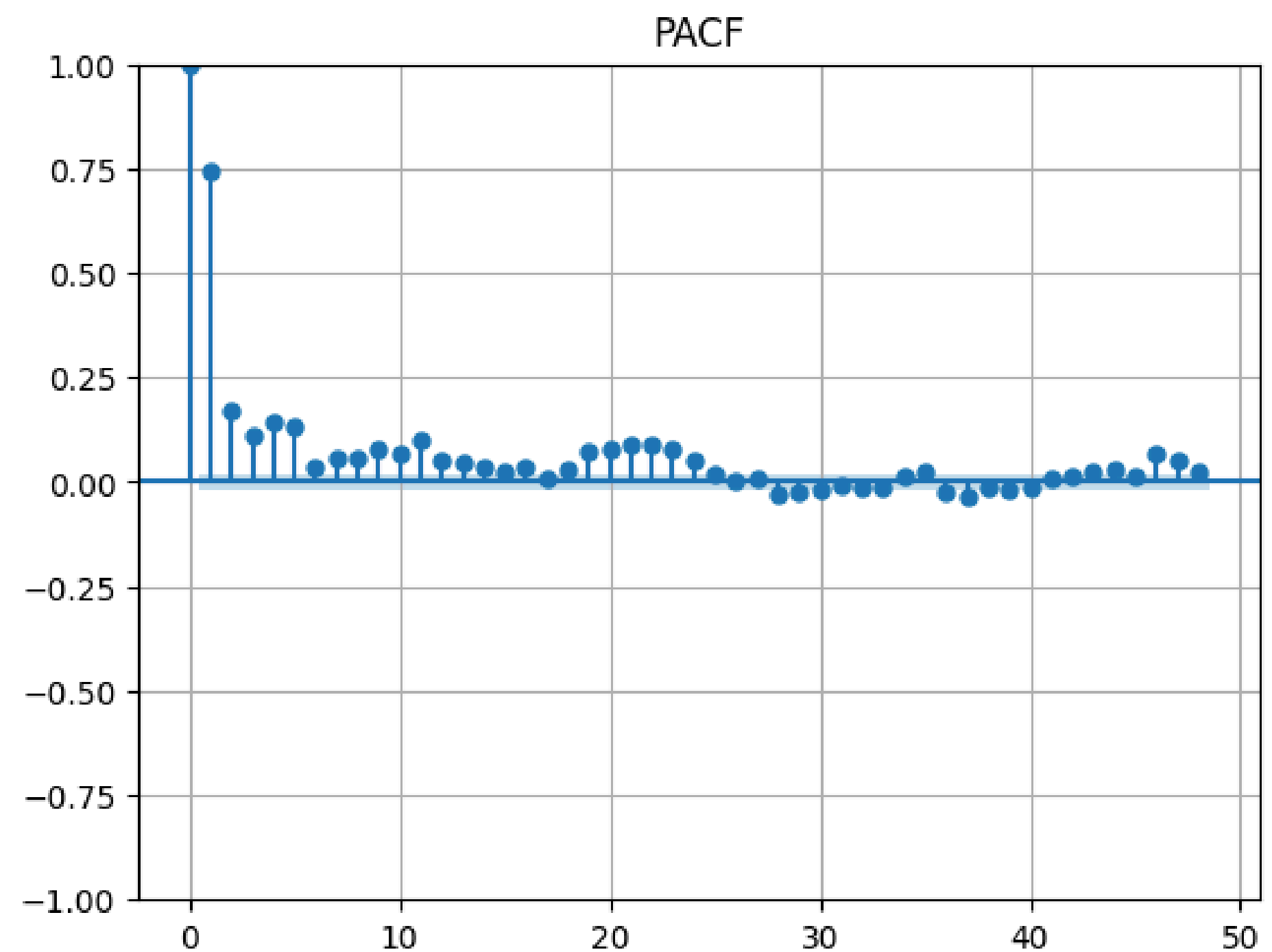
=== ADF Test: Original Series ===
ADF Statistic: -6.298645185513371
p-value: 3.455608142286192e-08
1%: -3.430991646794122
5%: -2.8618235703087764
10%: -2.566920939332377

=> Differencing Order $d = 0$





Choose q based on ACF



Choose p based on PACF

```

=== ARIMA Model Ranking ===
      Model      AIC      BIC      RMSE_Val
9  ARIMA(2,0,1) 103013.614812 103048.969054 111.415669
13 ARIMA(3,0,1) 103015.475405 103057.900495 111.483422
14 ARIMA(3,0,2) 103016.327784 103065.823723 111.657907
15 ARIMA(3,0,3) 103001.398014 103057.964801 112.079956
11 ARIMA(2,0,3) 102999.347930 103048.843869 112.191566
7  ARIMA(1,0,3) 103009.780094 103052.205184 113.012636
6  ARIMA(1,0,2) 103261.104114 103296.458356 116.695745
5  ARIMA(1,0,1) 103726.677390 103754.960784 120.756983
12 ARIMA(3,0,0) 103756.585119 103791.939361 120.882981
8  ARIMA(2,0,0) 103862.092709 103890.376102 121.037988
4  ARIMA(1,0,0) 104080.332940 104101.545486 121.183447
3  ARIMA(0,0,3) 104894.083213 104929.437455 121.486009
2  ARIMA(0,0,2) 105461.156460 105489.439853 121.529677
1  ARIMA(0,0,1) 107003.184797 107024.397342 121.556511
0  ARIMA(0,0,0) 111049.092261 111063.233957 121.650871
10 ARIMA(2,0,2) 104790.041512 104832.466602 122.475850

```

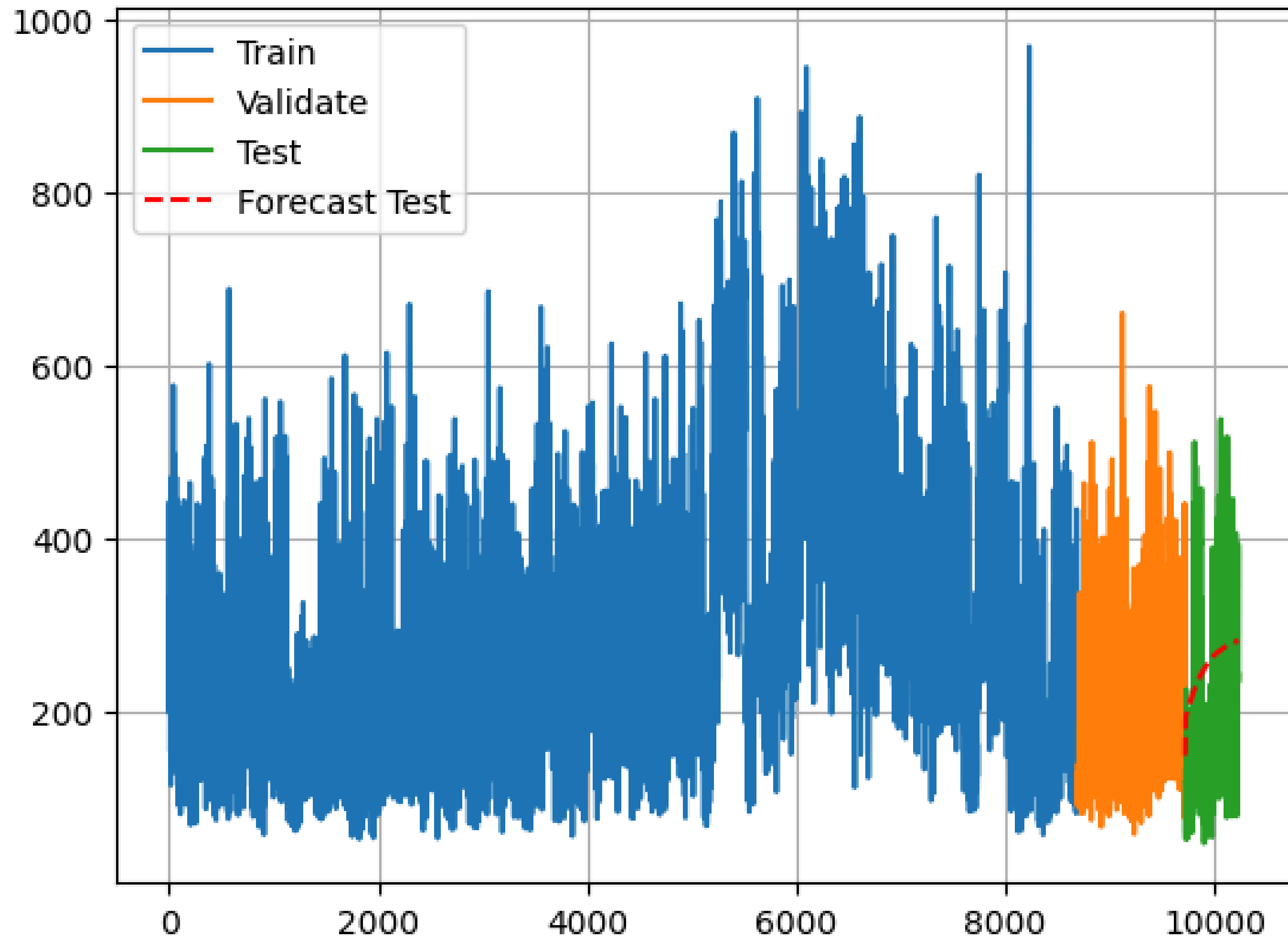
```

=== Optimal ARIMA Model ===
Model      ARIMA(2,0,1)
AIC        103013.614812
BIC        103048.969054
RMSE_Val   111.415669

```

**Choose the best
ARIMA Model**

ARIMA(2,0,1) - Predict Test



ARIMAX

Grid-search ARIMAX

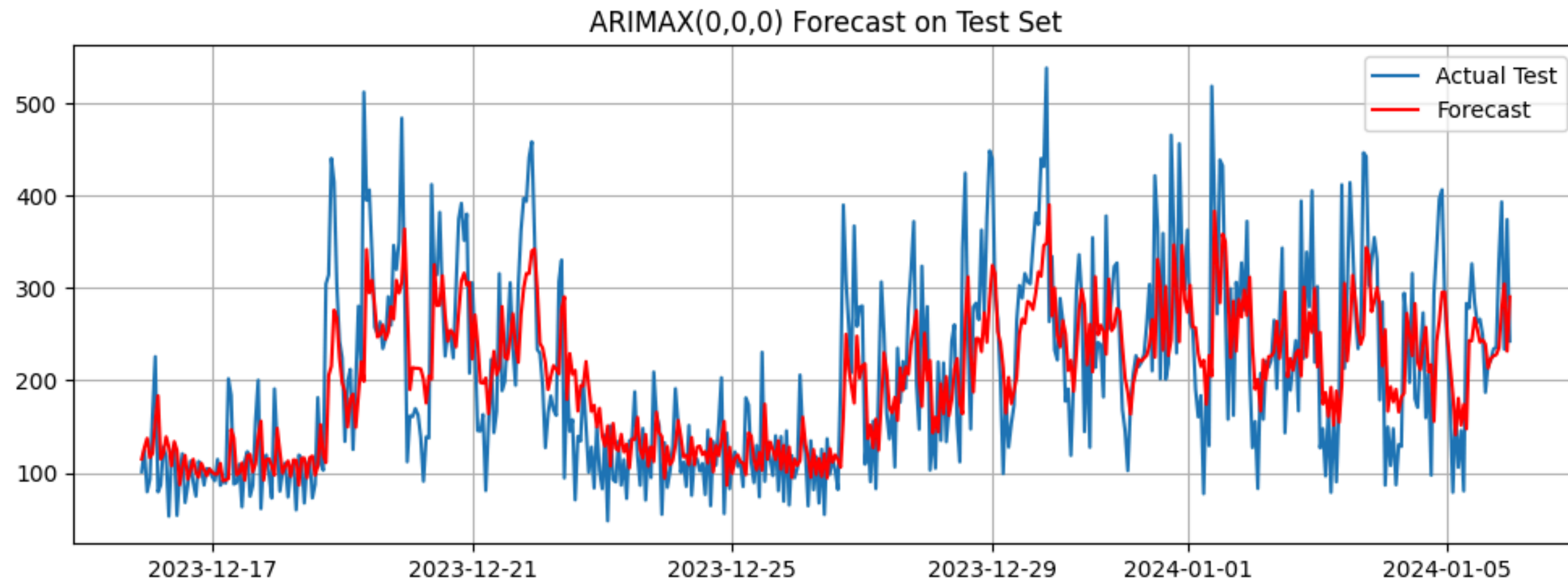
ARIMAX(0,0,0)	→ AIC=101322.36, MAE=57.08, RMSE=75.06
ARIMAX(0,0,1)	→ AIC=101283.47, MAE=58.11, RMSE=75.51
ARIMAX(0,0,2)	→ AIC=100920.86, MAE=94.86, RMSE=119.27
ARIMAX(0,0,3)	→ AIC=100502.73, MAE=103.56, RMSE=130.66
ARIMAX(1,0,0)	→ AIC=101006.12, MAE=70.27, RMSE=88.41
ARIMAX(1,0,1)	→ AIC=100524.70, MAE=104.75, RMSE=132.45
ARIMAX(1,0,2)	→ AIC=100514.80, MAE=104.57, RMSE=132.12
ARIMAX(1,0,3)	→ AIC=100497.87, MAE=104.25, RMSE=131.89
ARIMAX(2,0,0)	→ AIC=100826.81, MAE=81.05, RMSE=101.84
ARIMAX(2,0,1)	→ AIC=100573.22, MAE=103.76, RMSE=132.22
ARIMAX(2,0,2)	→ AIC=101305.85, MAE=57.11, RMSE=75.08
ARIMAX(2,0,3)	→ AIC=100505.54, MAE=103.64, RMSE=130.95
ARIMAX(3,0,0)	→ AIC=100684.79, MAE=88.82, RMSE=112.06
ARIMAX(3,0,1)	→ AIC=100506.77, MAE=104.56, RMSE=132.11
ARIMAX(3,0,2)	→ AIC=101051.44, MAE=71.83, RMSE=89.55
ARIMAX(3,0,3)	→ AIC=100493.59, MAE=104.03, RMSE=131.53

➔ **Best ARIMAX**
ARIMAX(0,0,0)

➔ This model achieves the lowest RMSE, meaning exogenous features explain most of the variance

➔ The weather variables and lag features already explain almost all changes in electricity consumption

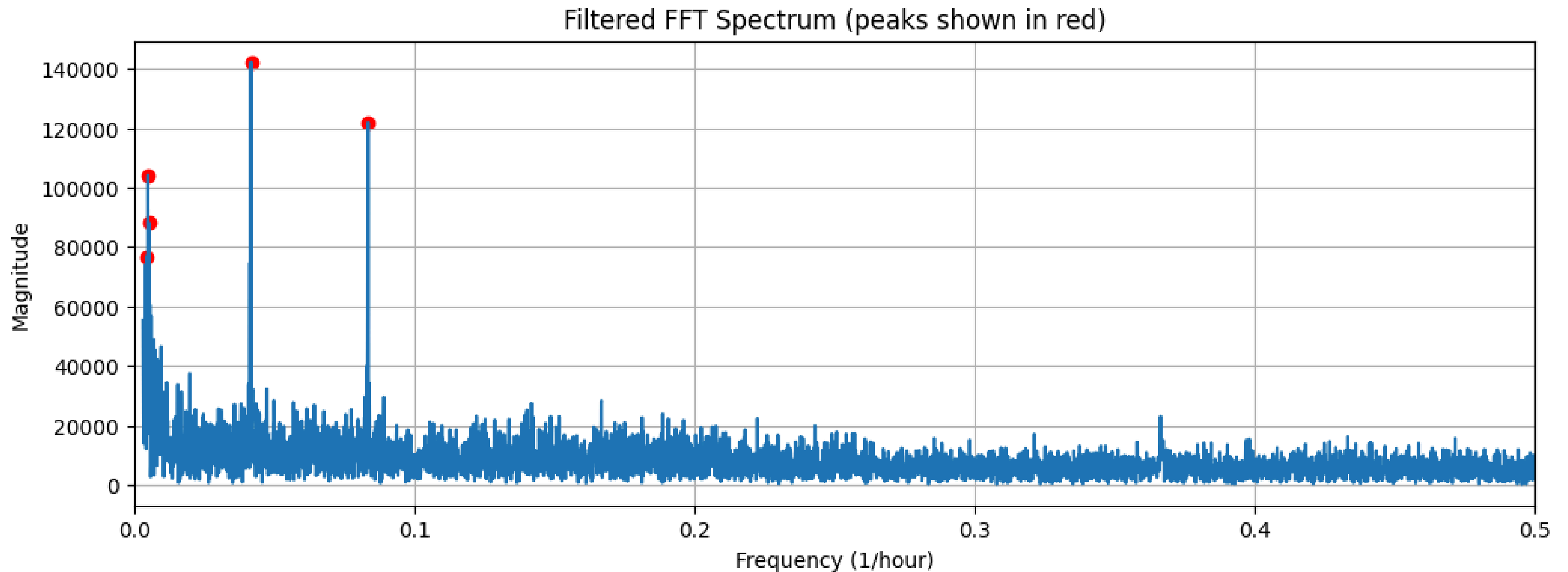
ARIMAX



MAE = 56.78
RMSE = 75.03
MAPE = 33.52%

- ➔ ARIMAX has improved over ARIMA by adding weather and lag factors.
- ➔ ARIMAX captures the overall trend well, but it cannot fully represent repeating daily consumption patterns.

SARIMAX



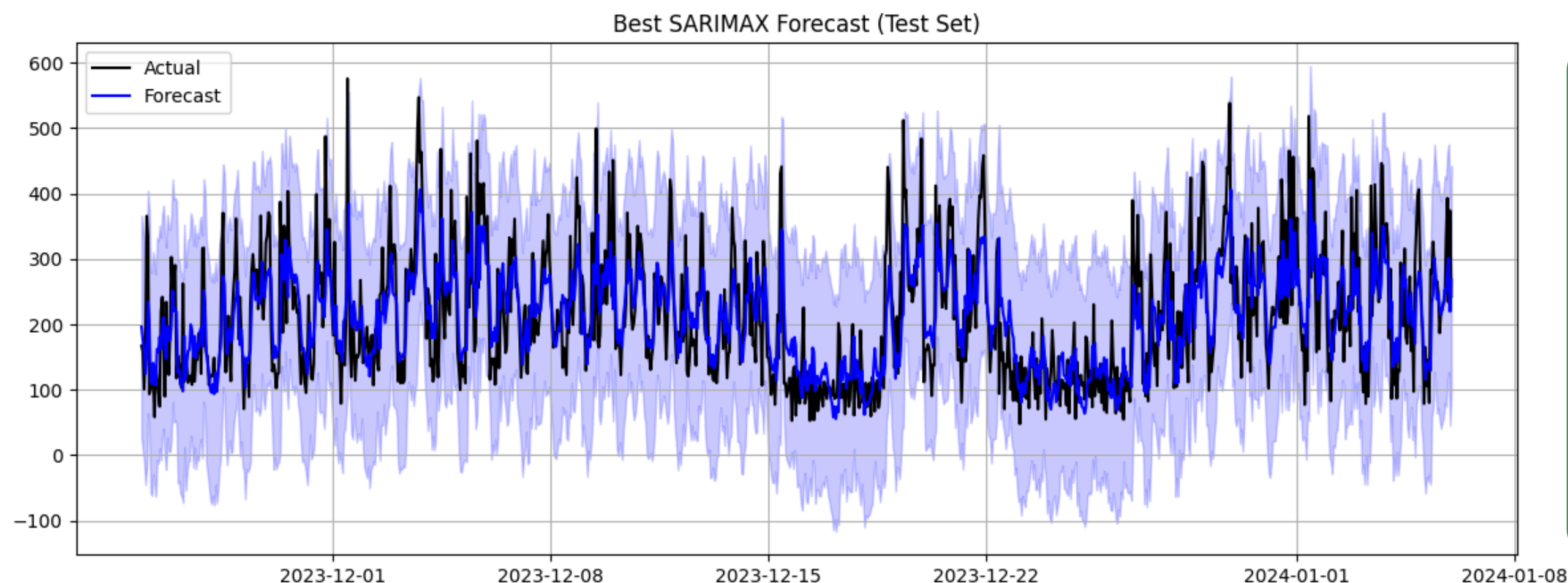
➡ Should choose $S = 24$ because strong peak $\sim 24h$.

SARIMAX

```
=== Running FAST SARIMAX Grid-search (16 models) ===  
SARIMAX(0, 0, 0)x(0, 1, 0, 24) → RMSE=129.37, AIC=98403.0  
SARIMAX(0, 0, 0)x(0, 1, 1, 24) → RMSE=77.03, AIC=94858.3  
SARIMAX(0, 0, 0)x(1, 1, 0, 24) → RMSE=107.63, AIC=97330.3  
SARIMAX(0, 0, 0)x(1, 1, 1, 24) → RMSE=77.83, AIC=94852.1  
SARIMAX(0, 0, 1)x(0, 1, 0, 24) → RMSE=135.63, AIC=98029.0  
SARIMAX(0, 0, 1)x(0, 1, 1, 24) → RMSE=77.16, AIC=94798.4  
SARIMAX(0, 0, 1)x(1, 1, 0, 24) → RMSE=110.93, AIC=97202.5  
SARIMAX(0, 0, 1)x(1, 1, 1, 24) → RMSE=79.91, AIC=94782.3  
SARIMAX(1, 0, 0)x(0, 1, 0, 24) → RMSE=138.90, AIC=97783.2  
SARIMAX(1, 0, 0)x(0, 1, 1, 24) → RMSE=85.01, AIC=94538.5  
SARIMAX(1, 0, 0)x(1, 1, 0, 24) → RMSE=116.75, AIC=96951.5  
SARIMAX(1, 0, 0)x(1, 1, 1, 24) → RMSE=88.64, AIC=94498.1  
SARIMAX(1, 0, 1)x(0, 1, 0, 24) → RMSE=141.19, AIC=97750.2  
SARIMAX(1, 0, 1)x(0, 1, 1, 24) → RMSE=121.93, AIC=94095.9  
SARIMAX(1, 0, 1)x(1, 1, 0, 24) → RMSE=142.97, AIC=96748.7  
SARIMAX(1, 0, 1)x(1, 1, 1, 24) → RMSE=145.73, AIC=94193.1
```

- ➔ **Best model**
SARIMAX(0,0,0) × (0,1,1,24)
- ➔ **This model is suitable due to its simple structure, strong Seasonal MA(1), which is suitable for 24h data, and has a stable performance**

SARIMAX



MAE = 54.37
RMSE = 71.23
MAPE = 30.23%

- ➡ SARIMAX outperforms ARIMAX and ARIMA by capturing 24-hour seasonality
- ➡ When there is a change in temperature, the model maintains an accurate trend due to the seasonal component and electricity lag

RANDOM FOREST

Using MAE with TimeSeriesSplit on the parameter grid:

'n_estimators':[50,100,150]

'max_depth':[10,20,None]

'min_samples_leaf':[1,2,3]

The best model selected is the

Random Forest Regressor

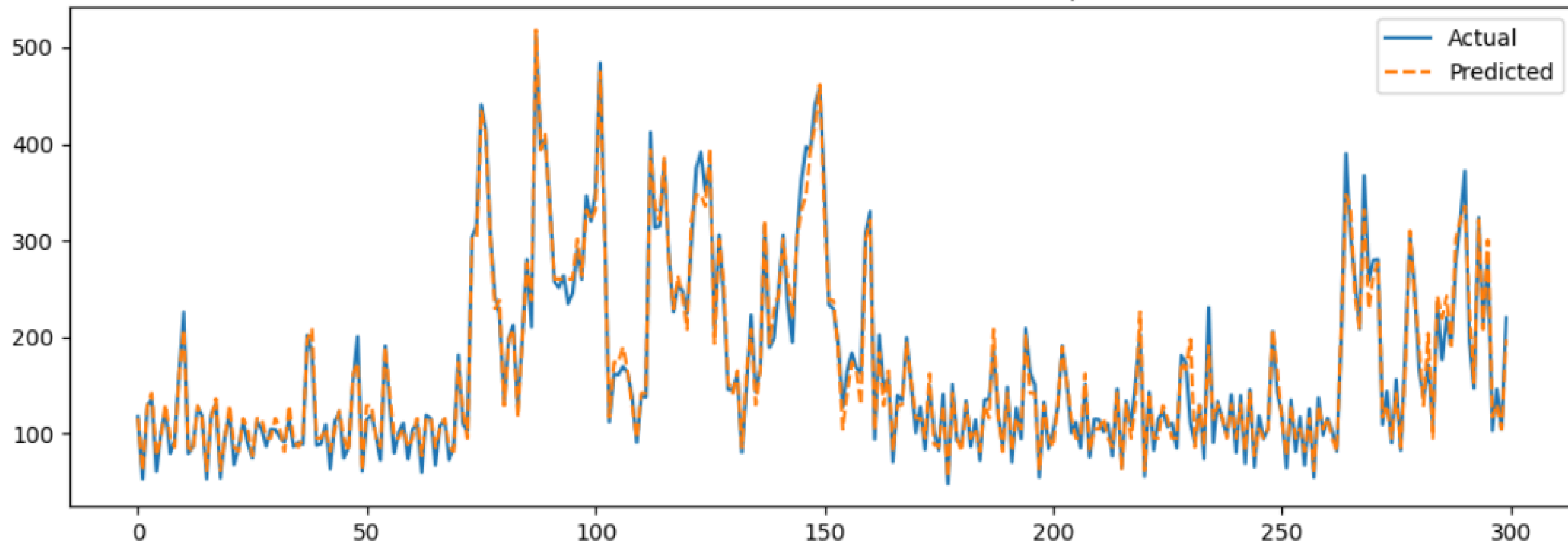
(n_estimators=150, max_depth=None, min_samples_leaf=2).

23	{'max_depth': None, 'min_samples_leaf': 2, 'n_...	5.796980	1.083060	3.141618	0.998773
14	{'max_depth': 20, 'min_samples_leaf': 2, 'n_es...	5.797013	1.083167	3.141588	0.998773
20	{'max_depth': None, 'min_samples_leaf': 1, 'n_...	5.800302	1.087938	3.154383	0.998763
19	{'max_depth': None, 'min_samples_leaf': 1, 'n_...	5.810432	1.089771	3.181780	0.998742
22	{'max_depth': None, 'min_samples_leaf': 2, 'n_...	5.817631	1.088140	3.168144	0.998753
13	{'max_depth': 20, 'min_samples_leaf': 2, 'n_es...	5.817695	1.088333	3.168286	0.998752
11	{'max_depth': 20, 'min_samples_leaf': 1, 'n_es...	5.818293	1.088056	3.148395	0.998768
10	{'max_depth': 20, 'min_samples_leaf': 1, 'n_es...	5.828582	1.090219	3.182559	0.998741
21	{'max_depth': None, 'min_samples_leaf': 2, 'n_...	5.847964	1.106665	3.251450	0.998686
12	{'max_depth': 20, 'min_samples_leaf': 2, 'n_es...	5.848232	1.106840	3.252670	0.998685
2	{'max_depth': 10, 'min_samples_leaf': 1, 'n_es...	5.851446	1.177081	3.310248	0.998638
5	{'max_depth': 10, 'min_samples_leaf': 2, 'n_es...	5.868778	1.170694	3.299650	0.998647
4	{'max_depth': 10, 'min_samples_leaf': 2, 'n_es...	5.872827	1.179342	3.334318	0.998618

RANDOM FOREST

```
mae_test = 3.029620644811419  
mape_test = 2.2495157535923327  
rmse_test = 4.653532444604543  
r2_test = 0.9980140542574952
```

Actual vs Predicted — First 300 Test Samples



DECISION TREE

	params	MAE
0	{'max_depth': 8, 'min_samples_leaf': 4}	14.686997
1	{'max_depth': 8, 'min_samples_leaf': 2}	14.791631
2	{'max_depth': 8, 'min_samples_leaf': 1}	14.956157
3	{'max_depth': 5, 'min_samples_leaf': 2}	15.058111
4	{'max_depth': 5, 'min_samples_leaf': 4}	15.058949
5	{'max_depth': 5, 'min_samples_leaf': 1}	15.073765
6	{'max_depth': None, 'min_samples_leaf': 4}	15.966052
7	{'max_depth': None, 'min_samples_leaf': 2}	16.874245
8	{'max_depth': None, 'min_samples_leaf': 1}	17.498842

With the param_grid:

'max_depth': [5, 8, None]

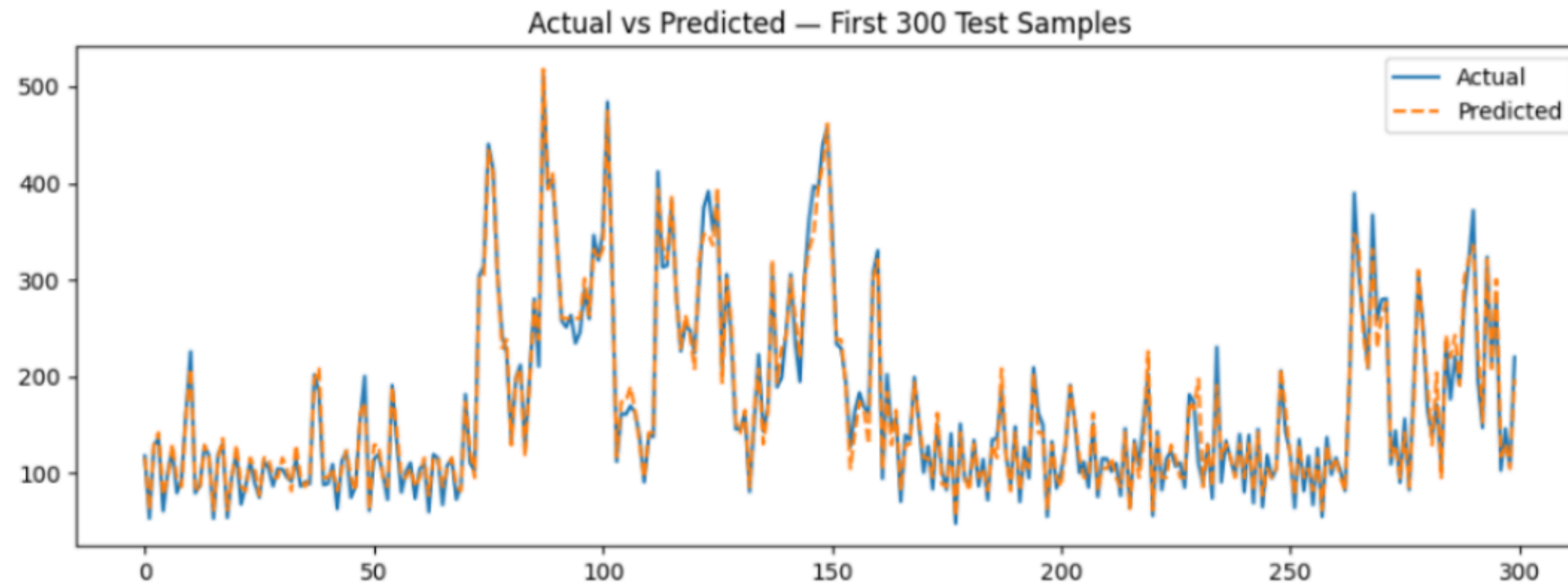
'min_samples_leaf': [1, 2, 4]

➡ Best option is Decision Tree with max_depth=8 and min_samples_leaf=4

DECISION TREE

Result for training:

```
MAE : 11.949476557431444  
RMSE: 15.601126973512889  
R2 : 0.9776789906352272  
MAPE: 6.924524436124689
```



What is LSTM model?

605224	1/5/2024 23:22	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	9.81	9.2	8.07	9.81	1011	46	1.74	158	12.58	10.96
605225	1/5/2024 23:23	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.35	10.71	8.09	13.99	1012	41	1.74	158	12.58	10.96
605226	1/5/2024 23:24	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.35	10.71	8.09	13.99	1012	41	1.74	158	12.58	10.96
605227	1/5/2024 23:25	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.32	10.68	8.07	13.96	1012	41	1.74	158	12.58	10.96
605228	1/5/2024 23:26	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.35	10.71	8.09	13.99	1012	41	1.74	158	12.58	10.96
605229	1/5/2024 23:27	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.32	10.68	8.07	13.96	1012	41	1.74	158	12.55	10.93
605230	1/5/2024 23:28	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.35	10.71	8.09	13.99	1012	41	1.74	158	12.58	10.96
605231	1/5/2024 23:29	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.32	10.68	8.07	13.96	1012	41	1.74	158	12.55	10.93
605232	1/5/2024 23:30	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.35	10.71	8.09	13.99	1012	41	1.74	158	15.1	13.89
605233	1/5/2024 23:31	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.32	10.68	8.07	13.96	1012	41	1.74	158	15.1	13.89
605234	1/5/2024 23:32	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.35	10.71	8.09	13.99	1012	41	1.74	158	15.1	13.89
605235	1/5/2024 23:33	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.32	10.68	8.07	13.96	1012	41	1.74	158	15.1	13.89
605236	1/5/2024 23:34	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.13	10.44	8.09	13.99	1012	40	1.73	159	15.1	13.89
605237	1/5/2024 23:35	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.1	10.41	8.07	13.96	1012	40	1.93	155	15.1	13.89
605238	1/5/2024 23:36	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.13	10.44	8.09	13.99	1012	40	1.73	159	15.1	13.89
605239	1/5/2024 23:37	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.1	10.41	8.07	13.96	1012	40	1.93	155	15.1	13.89
605240	1/5/2024 23:38	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.1	10.41	8.07	13.96	1012	40	1.93	155	15.1	13.89
605241	1/5/2024 23:39	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.13	10.44	8.09	13.99	1012	40	1.73	159	15.1	13.89
605242	1/5/2024 23:40	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.13	10.44	8.09	13.99	1012	40	1.73	159	15.1	13.89
605243	1/5/2024 23:41	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.1	10.41	8.07	13.96	1012	40	1.93	155	15.1	13.89
605244	1/5/2024 23:42	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.1	10.41	8.07	13.96	1012	40	1.93	155	15.1	13.89
605245	1/5/2024 23:43	159.8	1.75	129	159.46	225.75	0.7079	Clouds	scattered	12.1	10.41	8.07	13.96	1012	40	1.93	155	15.1	13.89
605246	1/5/2024 23:44	159.8	1.75	129	159.46	225.75	0.7079	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.99	12.75
605247	1/5/2024 23:45	159.8	1.75	129	159.46	225.75	0.7079	Clouds	few cloud:	12.35	10.68	8.09	13.99	1012	40	2.05	160	13.96	12.72
605248	1/5/2024 23:46	159.8	1.75	129	159.46	225.75	0.7079	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.96	12.72
605249	1/5/2024 23:47	159.8	1.75	129	159.46	225.75	0.7079	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	15.1	13.89
605250	1/5/2024 23:48	159.8	1.75	129	159.46	225.75	0.7079	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.96	12.72
605251	1/5/2024 23:49	159.8	1.75	129	159.46	225.75	0.7079	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.99	12.75
605252	1/5/2024 23:50	148.7	1.35	129.9	92.97	175.37	0.8479	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.99	12.75
605253	1/5/2024 23:51	148.7	1.35	129.9	92.97	175.37	0.8479	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.96	12.72
605254	1/5/2024 23:52	148.7	1.35	129.9	92.97	175.37	0.8479	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.99	12.75
605255	1/5/2024 23:53	316.3	2.53	129.1	81.46	326.62	0.9684	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.96	12.72
605256	1/5/2024 23:54	316.3	2.53	129.1	81.46	326.62	0.9684	Clouds	few cloud:	12.76	11.11	8.07	13.96	1012	39	2.05	160	13.99	12.75
605257	1/5/2024 23:55	316.3	2.53	129.1	81.46	326.62	0.9684	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.99	12.75
605258	1/5/2024 23:56	1526.3	13.58	124.8	736.67	1694.78	0.9006	Clouds	few cloud:	12.76	11.11	8.07	13.96	1012	39	2.05	160	13.99	12.75
605259	1/5/2024 23:57	1528.7	13.56	124.9	729.04	1693.64	0.9026	Clouds	few cloud:	12.32	10.65	8.07	13.96	1012	40	2.05	160	13.96	12.72
605260	1/5/2024 23:58	283.4	2.26	129.6	73.99	292.9	0.9676	Clouds	few cloud:	12.76	11.11	8.07	13.96	1012	39	2.05	160	13.99	12.75
605261	1/5/2024 23:59	283.4	2.26	129.6	73.99	292.9	0.9676	Clouds	few cloud:	12.76	11.11	8.07	13.96	1012	39	2.05	160	13.96	12.72

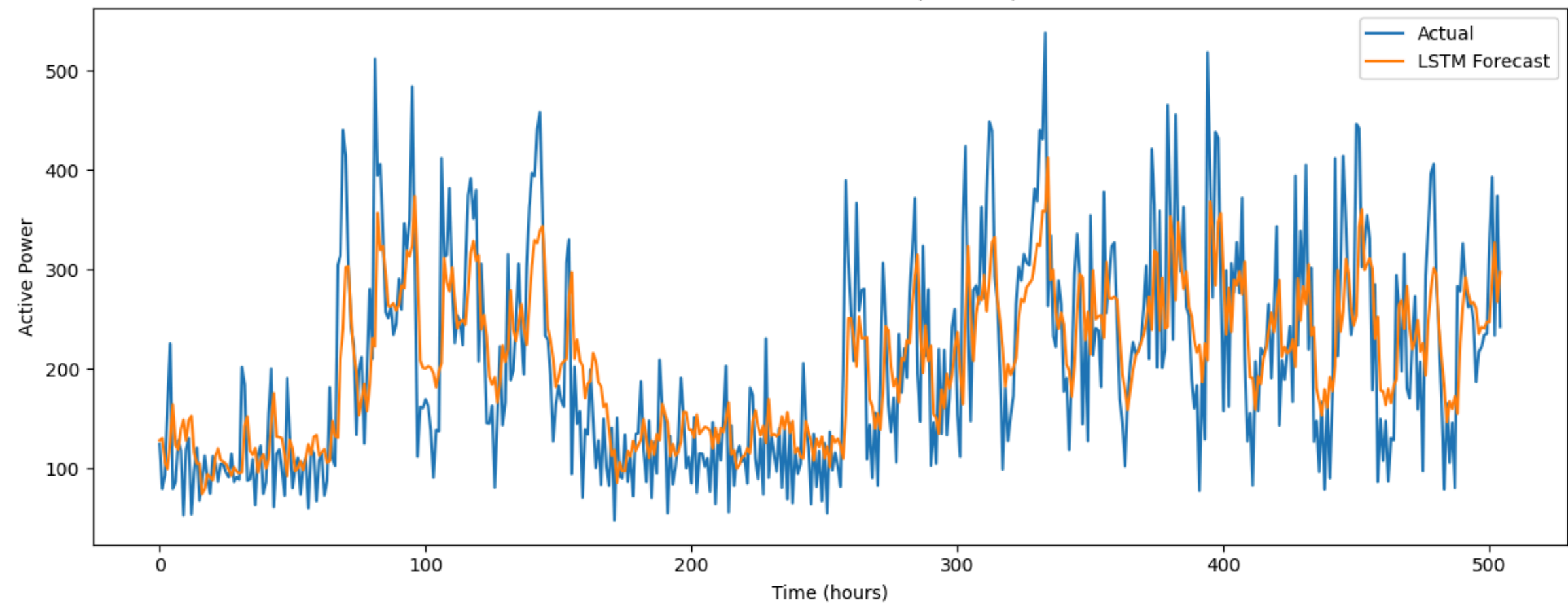
LSTM

Total sequences: 10088

Train: (8574, 24, 8) Val: (1009, 24, 8) Test: (505, 24, 8)

LSTM (1-step) - MAE: 55.2311, MSE: 5315.3831, RMSE: 72.9067, MAPE: 33.60%

LSTM Forecast vs Actual (Test Set)



Model Comparison

TABLE II
COMPARISON OF FORECASTING MODELS ON THE HOUSEHOLD ELECTRICITY DATASET

Metric	Statistical Models				Machine Learning / Deep Learning				Hybrid
	LinReg	ARIMA	ARIMAX	SARIMAX	DT	RF	XGB	LSTM	SARIMAX-LSTMX
MAE	60.55	95.84	56.78	54.37	11.95	3.03	56.66	55.80	84.69
RMSE	77.88	110.96	75.03	71.23	15.60	4.65	74.94	73.31	96.31
MAPE (%)	39.02	73.37	33.52	30.23	6.93	2.25	31.06	33.06	48.31
R ²	—	—	—	—	0.978	0.9980	0.3987	—	-2.913

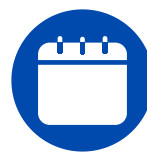
CONCLUSION

Key Insights from the Study

Weather Impact & Temperature

-  MAPE Reduction: 73% (ARIMA) → 30% (SARIMAX)
-  Temperature: Most significant factor
-  Hot days = Higher cooling demand

Daily & Temporal Patterns

-  Peaks: Noon & Evening
-  Clear Difference: Weekdays vs Weekends
-  Reflects household routines

Data Resampling Efficiency

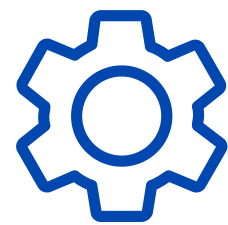
-  Minute → Hour conversion
-  Reveals key relationships (weather, habits)
-  Improve stability & prediction

Research Limitations



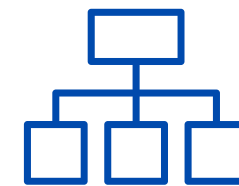
Data Limitations

- Dataset from one household → limited generalization
- Missing socioeconomic factors (members, income, appliance usage)
- Weather from OpenWeather API → may differ from actual conditions



Processing Limitations

- Resampling minute → hour removes short-term fluctuations



Modeling Limitations

- Focus on basic ML/statistical models
- No advanced models (Transformers)
- Feature selection not optimized



Evaluation Limitations

- No long-term model stability analysis
- Incomplete stationarity/structural tests

Recommendations



Regulators

- Adjust electricity supply by season & hot days
- Apply Time-of-Use pricing to reduce peak load



Electric Company

- Use Random Forest for short-term hourly forecasting
- Integrate predictive models into smart grids for anomaly detection & load optimization



Households

- Monitor consumption with prediction tools
- Detect unusual usage & adjust habits to save electricity

Thanks for listening